

Part V — Waves, Sound, and Light · Chapter 24

Wave Properties

How oscillations travel through space and carry energy across a medium

Drop a pebble into still water and watch the rings expand outward. The water at any one point does not travel to the shore; it merely bobs up and down as the disturbance passes through. Yet the disturbance itself — the pattern of crests and troughs — moves outward at a definite speed, delivering energy to the shore. This is a wave. Pluck a guitar string, and a similar thing happens in air: the string pushes air molecules, which push their neighbors, carrying a pattern of pressure variations outward at the speed of sound. The screen you are reading these words on was lit by light waves that left a lamp or the Sun and traveled through empty space at 300 000 km/s. In each case, the underlying physics is identical: an oscillation propagates, carrying energy from place to place. This chapter establishes the universal properties shared by all waves before the next two chapters specialize into sound and light.

24.1 What a Wave Is

A wave is a traveling disturbance that carries energy from one place to another without permanently transporting matter. This definition has three parts, each important.

Traveling disturbance: a wave is not a static pattern. It moves. The disturbance at one location propagates to neighboring locations over time, creating a moving pattern of some physical quantity (displacement, pressure, electric field, etc.).

Carries energy: waves transport energy from their source to distant locations. This is not incidental — it is the defining function of a wave in physics and engineering. The energy in sunlight that warms the Earth traveled across 150 million kilometers of space as electromagnetic waves. The energy that shatters a wine glass traveled from a speaker as sound waves.

Without permanently transporting matter: this distinguishes waves from particle flow. In a water wave, individual water molecules move in small circles and return to their original positions — they do not travel across the ocean. In a sound wave, air molecules vibrate back and forth but do not stream outward. The pattern moves; the medium does not.

Key Point: Exceptions exist: ocean currents do transport water. But the wave pattern on the surface is separate from the underlying current. It is the wave pattern — not the current — that carries the energy toward the shore.

24.2 Waves vs. Oscillations

Chapter 23 described oscillations — local, repetitive motion of a system around an equilibrium. Waves are what happen when that oscillation propagates. The distinction is critical:

An oscillation is a property of a single location or object: a mass on a spring, a pendulum, a vibrating string element. It occurs at a point in space and involves the same material moving back and forth repeatedly.

A wave is a spatial phenomenon: it involves many locations, each oscillating, with the phase of the oscillation shifting from one location to the next. The wave pattern is the set of all these oscillations arranged in space and time.

When you pluck a guitar string, you create a local disturbance. That disturbance travels along the string — the string element you plucked starts oscillating, then the adjacent element starts oscillating (slightly later), then the next, and so on. The wave is the propagating collection of all these oscillations, staggered in phase. At any instant, the string forms a spatial pattern of displacements. Over time, this pattern moves.

The Rule: Oscillation: one location, repeating in time. Wave: many locations, each oscillating, with the oscillation propagating from one to the next in space and time.

24.3 Mechanical vs. Electromagnetic Waves

All waves fall into two broad categories: mechanical waves and electromagnetic waves.

Mechanical waves require a physical medium — a material substance through which the oscillation propagates. Sound waves travel through air, water, and solid rock, but cannot travel through a vacuum. Seismic waves travel through the Earth. Water waves travel along the water surface. The medium provides the restoring force and inertia that sustain the oscillation and allow it to propagate. Remove the medium, and the wave cannot exist.

Electromagnetic waves are oscillations of electric and magnetic fields that require no medium. They can propagate through a perfect vacuum. Light, radio waves, microwaves, infrared, ultraviolet, X-rays, and gamma rays are all electromagnetic waves, differing only in frequency. They all travel through a vacuum at the same speed: $c = 3.00 \times 10^8$

m/s. This speed is one of the fundamental constants of nature.

Type	Requires Medium?	Speed in Vacuum	Examples
Mechanical	Yes	N/A (no vacuum propagation)	Sound, water waves, seismic
Electromagnetic	No	$c = 3.00 \times 10^8$ m/s	Light, radio, X-rays, microwaves

Table 24.1 *The two fundamental categories of wave.*

Historical Note: Heinrich Hertz demonstrated electromagnetic waves in 1887, confirming Maxwell's theoretical prediction of 1865. He generated radio waves in his laboratory and showed they reflected, refracted, and interfered just like light — proving they were waves of the same family. Hertz died at 36 from bone cancer; the unit of frequency (Hz) is named in his honor.

24.4 Transverse and Longitudinal Waves

Within mechanical waves, there is a second important classification based on the relationship between the direction of oscillation and the direction of wave propagation.

In a transverse wave, the medium oscillates perpendicular to the direction the wave travels. A wave on a taut string is transverse: the string oscillates up and down while the wave pattern moves horizontally along the string. Electromagnetic waves are also transverse: the electric and magnetic fields oscillate perpendicular to the direction of propagation.

In a longitudinal wave, the medium oscillates parallel to the direction the wave travels. The oscillations are compressions and rarefactions along the wave's path. Sound waves in air are longitudinal: air molecules oscillate back and forth in the same direction the sound travels. A slinky pushed at one end creates a longitudinal wave as compressions travel along its length.

Some wave types have both transverse and longitudinal components. Surface waves on water are an example: water particles move in ellipses that combine horizontal (longitudinal) and vertical (transverse) motion. Seismic S-waves (secondary waves) are transverse; seismic P-waves (primary waves) are longitudinal. The different speeds and behavior of these wave types allow seismologists to determine the location and depth of earthquakes.

Wave Type	Oscillation Direction	Examples
Transverse	Perpendicular to propagation	String waves, light, S-seismic waves
Longitudinal	Parallel to propagation (compression)	Sound, P-seismic waves, spring (slinky)
Mixed	Both transverse and longitudinal	Ocean surface waves

Table 24.2 *Classification of mechanical waves by direction of oscillation.*

24.5 Amplitude, Wavelength, Frequency, and Period

A traveling wave is described by four key quantities. Three of these — amplitude, frequency, and period — are shared with oscillations (Chapter 23). The fourth, wavelength, is unique to waves.

Amplitude (A) is the maximum displacement of the medium from its equilibrium value. For a transverse string wave, it is the maximum vertical deflection. For a sound wave, it is the maximum pressure variation above or below atmospheric. Amplitude is set by the source and determines how much energy the wave carries (energy is proportional to A^2).

Wavelength (λ , lambda) is the spatial period of the wave — the distance over which the wave pattern repeats. It is measured from one crest to the next crest (or trough to trough, or any other matching pair of points one cycle apart). Wavelength is to space what period is to time: the length of one full cycle in the spatial domain.

Frequency (f) and period (T) have the same meanings as in Chapter 23: the number of complete oscillations per second and the time for one oscillation, respectively, measured at a fixed point in the medium as the wave passes. The reciprocal relationship $f = 1/T$ still holds.

Quantity	Symbol	Unit	Description
Amplitude	A	m	Maximum displacement from equilibrium
Wavelength	λ	m	Spatial length of one complete cycle
Frequency	f	Hz = s^{-1}	Oscillations per second at a fixed point

Period	T	s	Time for one complete oscillation
Wave speed	v	m/s	Speed at which the wave pattern advances

Table 24.3 *The five fundamental descriptors of a traveling wave.*

24.6 Wave Speed

Wave speed is the speed at which the wave pattern moves through the medium. It is distinct from the speed of the oscillating particles in the medium (which oscillate back and forth but do not travel with the wave). Wave speed is a property of the medium, not of the source or the amplitude.

The fundamental wave speed equation connects the three primary wave quantities:

$$v = f\lambda$$

Wave speed equals frequency times wavelength. This relation holds for all wave types — mechanical and electromagnetic alike.

This equation can be understood intuitively. In one second, f complete wave cycles pass a fixed point. Each cycle has a spatial length of λ . So the front of the wave has advanced $f \times \lambda$ meters per second.

For mechanical waves, wave speed depends on the properties of the medium: its stiffness (elasticity) and its inertia (density). Stiffer media transmit waves faster; denser media transmit them slower. For a string under tension T with linear mass density μ (mass per unit length), the transverse wave speed is $v = \sqrt{T/\mu}$. For sound in air, the speed depends on air temperature, pressure, and molecular mass.

For electromagnetic waves in a vacuum, the speed is always $c = 3.00 \times 10^8$ m/s regardless of frequency. This is one of the most important constants in physics. In a material medium, electromagnetic waves travel at $v = c/n$, where n is the index of refraction of the medium (always $n \geq 1$, so $v \leq c$).

Example — Wave speed calculation A wave on a string has frequency $f = 20.0$ Hz and wavelength $\lambda = 0.300$ m. Speed: $v = f\lambda = 20.0 \times 0.300 = 6.00$ m/s.

Example — Finding wavelength Sound travels at 343 m/s in air at 20°C. Middle C has frequency 261.6 Hz. Wavelength: $\lambda = v/f = 343/261.6 = 1.31$ m.

Study Tip: When solving wave problems: v is set by the medium; f is set by the source; λ is determined by both through $\lambda = v/f$. Changing the source frequency changes λ but not v (in a fixed medium).

24.7 Energy Transport in Waves

Waves carry energy from their source to distant locations. This is the primary function of waves in nature and technology. Understanding how much energy a wave carries and how that energy is distributed in space is central to wave physics.

The intensity of a wave is the power delivered per unit area perpendicular to the wave's direction of travel: $I = P/A$, measured in watts per square meter (W/m^2). Intensity is the key measurable quantity for energy transport.

For a wave spreading outward from a point source in three dimensions (like sound from a speaker, or light from a lamp), the wave energy spreads over a sphere whose area increases as $4\pi r^2$. Since total power is conserved, intensity decreases with the square of distance: $I \propto 1/r^2$. This is the inverse-square law for wave intensity. Doubling the distance from a source reduces the intensity to one-quarter; tripling the distance reduces it to one-ninth.

$$I \propto 1/r^2$$

$$I_1/I_2 = r_2^2/r_1^2$$

Inverse-square law for wave intensity from a point source. Valid for three-dimensional spreading (sound, light, radio).

Wave intensity is proportional to the square of the amplitude: $I \propto A^2$. This is the same relationship as for oscillator energy in Chapter 23. Physically, a wave with twice the amplitude delivers four times the energy per unit area per unit time.

For waves confined to two dimensions (like surface waves on water), energy spreads over a circle of circumference $2\pi r$, and intensity decreases as $1/r$. For waves confined to one dimension (like a wave on a string or in a tube), intensity does not decrease with distance in the ideal case (no damping).

24.8 Wave Fronts and Direction of Propagation

A wave front is a surface connecting all points in a medium that are at the same phase of oscillation — for example, the crest of a wave or the leading edge of a disturbance. Wave fronts propagate outward from the source at the wave speed.

For a point source in a uniform medium, wave fronts are concentric spheres (in three dimensions) or concentric circles (in two dimensions, such as on a water surface). These are spherical waves. The direction of propagation is always perpendicular to the wave front surface, pointing radially outward.

Far from a point source, the curved wave fronts become nearly flat over any small

region. These nearly flat wave fronts are called plane waves. Plane waves have parallel rays and constant amplitude across the wave front. Many physical waves can be treated as plane waves in the regions where they are analyzed — this is a useful simplification that greatly reduces the mathematical complexity of wave problems.

The ray is an imaginary line perpendicular to the wave front, indicating the direction of wave propagation and energy transport. Ray diagrams are used extensively in optics and acoustics to trace the paths of waves through systems with boundaries, mirrors, and lenses. The ray model is a simplification valid when the wavelength is much smaller than any obstacles or apertures the wave encounters.

24.9 Reflection of Waves

Reflection occurs when a wave encounters a boundary between two media and is at least partly redirected back into the original medium. All wave types — mechanical and electromagnetic — reflect from boundaries.

The law of reflection states that the angle of reflection equals the angle of incidence, both measured from the normal (perpendicular) to the reflecting surface. This law applies to all waves when the reflecting surface is smooth relative to the wavelength (specular reflection). When the surface is rough (bumpy on the scale of the wavelength), reflection is diffuse and scattered in many directions.

$$\theta_{\text{reflected}} = \theta_{\text{incident}} \quad (\text{Law of Reflection})$$

Both angles measured from the normal to the reflecting surface. This law holds for all wave types.

When a transverse wave on a string reflects from a fixed end (a wall), the reflected wave is inverted — a crest becomes a trough. This is a phase change of 180° on reflection from a denser medium. When the wave reflects from a free end (an end that can move), the reflected wave is not inverted. This distinction matters greatly for standing wave formation (Section 24.14).

Partial reflection and partial transmission occur at most real boundaries. When sound hits a wall, most is reflected and some is transmitted into the wall material. When light hits a glass surface, about 4% is reflected and 96% is transmitted. The fractions depend on the wave type and the properties of the two media.

24.10 Refraction of Waves

Refraction is the change in direction of a wave as it passes from one medium into another with a different wave speed. Because the wave speed changes at the boundary,

and the frequency remains constant (it is set by the source), the wavelength must change ($\lambda = v/f$). The change in wavelength is accompanied by a change in the wave's direction.

The direction change occurs because one part of the wave front reaches the boundary and slows down while the other part is still in the original medium and continues at the original speed. This creates a turning of the wave front, bending the direction of propagation toward the normal if the wave slows, or away from the normal if the wave speeds up.

Snell's Law quantifies refraction for light and, in the same form, for any wave crossing a planar boundary:

$$n_1 \sin\theta_1 = n_2 \sin\theta_2 \quad \text{or equivalently} \quad v_2 \sin\theta_1 = v_1 \sin\theta_2$$

Snell's Law of refraction. θ is measured from the normal. n is the index of refraction ($n = c/v$ for light). For mechanical waves, use speeds directly.

Refraction explains why a swimming pool looks shallower than it is (light bends as it exits the water), why mirages form over hot roads (air temperature creates a speed gradient), why sound can bend around hills (temperature and wind gradients cause refraction), and why lenses focus light (curved surfaces cause progressive refraction).

24.11 Diffraction of Waves

Diffraction is the bending and spreading of a wave as it passes through an opening or around an obstacle. It is a uniquely wave-like behavior — particles do not diffract. The observation of diffraction is therefore definitive evidence that a phenomenon is wave-like in nature.

The amount of diffraction depends on the ratio of the wavelength to the size of the opening or obstacle. When λ is much smaller than the opening, the wave passes through with minimal spreading (like light through a window). When λ is comparable to or larger than the opening, significant spreading occurs: the wave bends around the edges and creates a complex diffraction pattern beyond the opening.

This is why sounds (with wavelengths of centimeters to meters) diffract readily around buildings and corners, while visible light (with wavelengths of about 400–700 nm) does not noticeably diffract around everyday objects. You can hear someone around a corner, but you cannot see them, because sound wavelengths are comparable to building dimensions while light wavelengths are not.

Diffraction is exploited in X-ray crystallography, where X-rays (wavelength comparable

to atomic spacing) diffract from crystal lattices to produce patterns that reveal atomic structure. The double-helix structure of DNA was discovered using X-ray diffraction data collected by Rosalind Franklin.

The Rule: Significant diffraction occurs when $\lambda \approx d$, where d is the size of the obstacle or opening. When $\lambda \ll d$, wave behavior approximates ray behavior (straight-line propagation). When $\lambda \gg d$, extensive spreading occurs.

24.12 Interference of Waves

Interference is the combination of two or more waves at the same location in space and time. It is, along with diffraction, the most distinctive wave behavior — one that has no counterpart in particle motion.

When two waves meet, the resulting displacement at each point is the sum of the individual displacements (this is the superposition principle, Section 24.13). Interference can be constructive (the waves reinforce each other, producing larger amplitude) or destructive (the waves partially or completely cancel each other).

Constructive interference occurs when two waves arrive at a point in phase — that is, when their crests arrive simultaneously. The displacements add, producing amplitude equal to the sum of the individual amplitudes.

Destructive interference occurs when two waves arrive out of phase — when the crest of one coincides with the trough of the other. The displacements partially or completely cancel. For two waves of equal amplitude arriving exactly 180° out of phase, the result is zero displacement: complete cancellation.

$$\text{Constructive: } \delta = m\lambda \quad (m = 0, \pm 1, \pm 2, \dots)$$

$$\text{Destructive: } \delta = (m + \tfrac{1}{2})\lambda$$

δ = path length difference between two coherent sources. These conditions apply when both sources have the same frequency and a fixed phase relationship.

Interference is responsible for the iridescent colors in soap bubbles and oil films (thin-film interference), the noise-canceling effect in noise-canceling headphones (destructive interference of sound), the bright and dark bands in double-slit experiments (Young's experiment), and the operating principle of radio antennas, interferometric telescopes, and laser-based sensors.

24.13 Superposition Principle

The superposition principle states that when two or more waves overlap in a linear medium, the resultant displacement at any point at any time is the algebraic sum of the displacements of the individual waves. Each wave passes through the other unchanged; after the waves separate, each continues as if the interaction never occurred.

This principle is fundamental and far-reaching. It holds exactly for linear media — media in which the restoring force is proportional to displacement, including most real media for small amplitudes. It fails for very large amplitudes (nonlinear behavior), where waves can interact in more complex ways (a field called nonlinear optics or nonlinear acoustics).

Superposition underlies interference and all complex wave behavior. The shape of a wave on a guitar string at any instant is the superposition of all the normal modes (harmonics) simultaneously present. The signal in a telephone wire is the superposition of thousands of simultaneously transmitted conversations. The light from a fluorescent lamp is the superposition of many individual atomic emission frequencies. Fourier analysis — the mathematical decomposition of any periodic function into sine and cosine components — is the mathematical expression of superposition.

Bridge Note *Fourier analysis is the mathematical embodiment of superposition: any periodic wave can be decomposed into a sum of sinusoidal components at different frequencies and amplitudes. Fourier transforms, covered in university mathematics, are used throughout physics, engineering, signal processing, and image analysis. The concepts of constructive and destructive interference are the physical basis for understanding Fourier decomposition.*

24.14 Standing Waves (Introductory)

When a wave reflects from a boundary and returns along the same path, the original (incident) wave and the reflected wave coexist and superimpose. For specific frequencies, this superposition produces a standing wave: a pattern that appears stationary rather than traveling.

A standing wave has nodes — points that never move — and antinodes — points that oscillate with maximum amplitude. Nodes occur where the incident and reflected waves always cancel (destructive interference). Antinodes occur where they always reinforce (constructive interference). Unlike a traveling wave, a standing wave does not transport energy along the medium; instead, energy is stored in the oscillating pattern.

Standing waves form when the length of the medium fits a whole number of half-wavelengths. For a string fixed at both ends (length L), standing waves occur at frequencies:

$$f_n = n \times v / (2L) \quad (n = 1, 2, 3, \dots)$$

Resonant frequencies (harmonics) of a string fixed at both ends. $n = 1$ is the fundamental; $n = 2, 3, \dots$ are higher harmonics (overtones).

The lowest frequency ($n = 1$) is the fundamental frequency. Higher values of n give overtones or harmonics. Musical instruments produce standing waves: the vibrating string of a guitar, the air column in a flute, the membrane of a drum — all resonate at their characteristic harmonic frequencies. The combination of harmonics present and their relative amplitudes determines the timbre (tone quality) that distinguishes one instrument from another playing the same note.

Historical Note: Thomas Young performed his famous double-slit experiment in 1801, demonstrating that light produces interference fringes consistent with wave behavior. This was decisive evidence that light is a wave — contradicting Newton's corpuscular theory. Young's experiment is now reproduced routinely with electrons and even individual photons, revealing the quantum wave-particle duality at the heart of modern physics.

24.15 Common Mistakes in Wave Analysis

- Confusing wave speed with particle (medium) speed. Wave speed is the speed of the wave pattern. Medium particles oscillate back and forth but do not travel with the wave. These are completely different speeds.
- Thinking that a faster wave has higher frequency. In a fixed medium, wave speed is fixed. Higher frequency simply means shorter wavelength ($\lambda = v/f$). Both frequency and wavelength can vary independently of wave speed.
- Applying the inverse-square law to one-dimensional waves (waves on a string, in tubes). The inverse-square law applies only to waves spreading in three dimensions from a point source. Confined waves do not obey it.
- Confusing destructive interference with energy destruction. Destructive interference does not destroy energy; it redistributes it. Where two waves cancel, they reinforce elsewhere. Total energy is conserved.
- Forgetting that refraction changes direction and wavelength but not frequency. Frequency is set by the source and is unchanged as the wave crosses a boundary. Speed and wavelength both change.
- Treating all waves as transverse. Sound is longitudinal. Seismic P-waves are longitudinal. Not all mechanical waves involve up-and-down motion.
- Confusing standing waves with traveling waves. Standing waves have fixed nodes and antinodes. Traveling waves carry energy; standing waves store it. A standing wave is the superposition of two traveling waves moving in opposite directions.

Common Mistake: The wave speed formula $v = f\lambda$ holds for all traveling waves without exception. Always check that you have identified the correct wave speed for the medium before applying this formula.

Chapter Summary

A wave is a traveling disturbance that transports energy without permanently transporting matter. Wave behavior is universal across mechanical and electromagnetic types; the same equations and principles apply.

- Mechanical waves require a medium; electromagnetic waves do not and travel at $c = 3.00 \times 10^8$ m/s in vacuum.
- Transverse waves oscillate perpendicular to propagation; longitudinal waves oscillate parallel to propagation.
- Four descriptors: amplitude A , wavelength λ , frequency f , period T . Wave speed $v = f\lambda$ is the master relationship.
- Wave speed is a property of the medium; frequency is set by the source; wavelength is determined by both through $\lambda = v/f$.
- Intensity $I = P/A$; for spherical spreading, $I \propto 1/r^2$ (inverse-square law); $I \propto A^2$.
- Reflection: angle of reflection = angle of incidence. Refraction: wave bends when crossing a speed boundary (Snell's Law). Diffraction: waves spread when passing openings comparable in size to λ .
- Superposition: displacements of overlapping waves add algebraically. Constructive interference ($\delta = m\lambda$) adds; destructive interference ($\delta = (m + \frac{1}{2})\lambda$) cancels.
- Standing waves form when reflected and incident waves superimpose at resonant frequencies: $f_n = nv/(2L)$ for a fixed string.

Key Terms

wave: A traveling disturbance that transports energy through space or a medium without permanently transporting matter.

mechanical wave: A wave that requires a physical medium to propagate; examples include sound, water waves, and seismic waves.

electromagnetic wave: A wave consisting of oscillating electric and magnetic fields that propagates through a vacuum at $c = 3.00 \times 10^8$ m/s; includes light, radio, and X-rays.

transverse wave: A wave in which the medium oscillates perpendicular to the direction of propagation; examples include string waves and light.

longitudinal wave: A wave in which the medium oscillates parallel to the direction of propagation; sound is the primary example.

wavelength (λ): The spatial distance over which the wave pattern repeats; the distance between successive crests (or troughs); measured in meters.

wave speed (v): The speed at which the wave pattern advances through the medium; $v = f\lambda$; determined by the medium's properties.

intensity (I): Power delivered per unit area perpendicular to the wave direction; $I = P/A$; measured in W/m^2 ; proportional to A^2 .

inverse-square law: The relationship $I \propto 1/r^2$ describing how wave intensity decreases with distance from a point source spreading in three dimensions.

wave front: A surface connecting all points of the same phase in a wave; perpendicular to the direction of propagation.

ray: A line perpendicular to the wave front indicating the direction of wave propagation and energy transport.

reflection: The redirection of a wave back into its original medium when it encounters a boundary; angle of reflection = angle of incidence.

refraction: The change in direction of a wave as it crosses a boundary where wave speed changes; governed by Snell's Law.

diffraction: The bending and spreading of a wave around obstacles or through openings; most pronounced when $\lambda \approx$ opening size.

interference: The superposition of two or more waves; constructive when in phase (amplitudes add); destructive when out of phase (amplitudes cancel).

superposition principle: The principle that the resultant displacement of overlapping waves equals the algebraic sum of individual displacements.

standing wave: The stationary wave pattern produced by the superposition of two identical waves traveling in opposite directions; has fixed nodes and antinodes.

node: A point in a standing wave that remains stationary (zero amplitude); caused by consistent destructive interference.

antinode: A point in a standing wave that oscillates with maximum amplitude; caused by consistent constructive interference.

fundamental frequency: The lowest resonant frequency of a standing wave system; $n = 1$ in the harmonic series.

Concept Review Questions

Answer in complete sentences. No calculations required unless specified.

1. Explain the statement: "a wave transports energy without permanently transporting matter." Use a water wave as your example.
2. What is the difference between a transverse and a longitudinal wave? Classify the following: (a) sound in air, (b) light, (c) a wave on a guitar string, (d) seismic P-waves.
3. A loudspeaker vibrates at 500 Hz in air ($v = 343$ m/s) and in water ($v = 1480$ m/s). Which property changes when sound crosses from air to water: frequency, wavelength, or wave speed? Explain.
4. Explain why you can hear someone talking around a corner of a building but you cannot see them. What role does wavelength play in your answer?
5. Two identical speakers emit the same tone toward a listener. If one speaker is moved so the path length difference is 2.5 wavelengths, will the listener hear constructive or destructive interference? Explain.
6. What is the superposition principle? Why does it imply that destructive interference does not destroy energy?
7. Explain what nodes and antinodes are in a standing wave. Why do nodes remain stationary?
8. State the law of reflection and explain how it applies to (a) sound reflecting off a flat wall and (b) light reflecting off a mirror.

Wave Speed Problems

9. A wave on a string has wavelength 0.40 m and frequency 25 Hz. Find the wave speed.
10. Sound travels at 343 m/s in air. A sound wave has wavelength 0.686 m. Find its frequency and period.
11. Light has a frequency of 6.00×10^{10} Hz in vacuum. Find its wavelength.
12. Radio station WGBH broadcasts at 89.7 MHz. Find the wavelength of its radio waves in air.

13. A wave in water has frequency 2.0 Hz and wavelength 1.5 m. Find the wave speed. If the same frequency wave enters a region where the depth increases and the speed becomes 3.6 m/s, what is the new wavelength?
14. Light travels at 3.00×10^8 m/s in vacuum and at 2.26×10^8 m/s in glass. A light wave of frequency 5.00×10^{14} Hz: (a) Find its wavelength in vacuum. (b) Find its wavelength in glass. (c) Does the frequency change when it enters glass?

Wavelength/Frequency Exercises

15. Complete the table ($v = 343$ m/s for all):

Frequency (Hz)	Period (ms)	Wavelength (m)
100	?	?
?	2.0	?
4000	?	?
?	?	0.0857

16. A microwave oven operates at 2.45 GHz. Find the wavelength of the microwaves. Why must the oven's cavity dimensions relate to this wavelength?
17. Medical ultrasound uses frequencies from 2 MHz to 15 MHz. Speed of ultrasound in soft tissue is approximately 1540 m/s. Find the wavelength range.

Wave Behavior Analysis

18. Light strikes a glass surface at 30° from the normal ($n_{\text{air}} = 1.00$, $n_{\text{glass}} = 1.50$). Find the angle of refraction in the glass using Snell's Law.
19. Sound in air (343 m/s) strikes a water surface at 25° from the normal. Speed of sound in water is 1480 m/s. Find the angle of refraction in the water.
20. Two speakers separated by 3.00 m emit sound at 686 Hz ($v = 343$ m/s, $\lambda = 0.500$ m). A listener is directly in front of one speaker at 4.00 m distance. Find the path length difference to the two speakers and determine whether the listener hears constructive or destructive interference.

21. A wave with amplitude 0.020 m and intensity I_1 at distance 1.0 m from a point source. (a) Find the intensity at 3.0 m. (b) What amplitude corresponds to the intensity at 3.0 m?
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Superposition Problems

22. Two waves on a string have amplitudes 3.0 cm and 4.0 cm and arrive at a point in phase. What is the resultant amplitude? What would it be if they arrived exactly out of phase?
23. A guitar string of length 0.650 m produces standing waves at wave speed 420 m/s. Find the frequencies of the first four harmonics.
24. A string fixed at both ends has its fundamental frequency at 120 Hz. At what frequencies do the 2nd, 3rd, and 4th harmonics occur? What is the relationship between string length and wavelength for each?
25. Two speakers emit sound of wavelength 1.20 m. They are in phase. A listener walks perpendicular to the line joining the speakers. At what distances from the center does the listener experience (a) maximum (constructive) and (b) minimum (destructive) intensity? (Treat the problem as a simple path difference problem; speakers separated by 2.40 m, listener at 5.00 m perpendicular distance.)

Chapter 24 Checkpoint

Before moving on to Chapter 25, confirm you can do each of the following.

I can...	See Section
Define a wave and explain how it differs from particle motion	24.1
Distinguish oscillation from wave propagation	24.2
Compare mechanical and electromagnetic waves; state the speed of light in vacuum	24.3
Classify a wave as transverse or longitudinal given a description	24.4

Define amplitude, wavelength, frequency, period, and wave speed	24.5
Apply $v = f\lambda$ to find any one quantity given the other two	24.6
Apply the inverse-square law for intensity from a point source	24.7
Explain what a wave front and a ray represent	24.8
State the law of reflection and apply it to waves	24.9
Explain refraction using wave speed changes at a boundary; apply Snell's Law	24.10
Explain why diffraction is most pronounced when $\lambda \approx$ opening size	24.11
Distinguish constructive from destructive interference using path difference	24.12
State the superposition principle and its scope of validity	24.13
Calculate standing wave resonant frequencies for a fixed string	24.14
Identify and avoid common errors in wave problems	24.15

Continue to Chapter 25: Sound Waves and Resonance →